Introduction to Probability and Statistics

(EE 354 / CE 361)

Independence of DRVs

Today's Agenda

- Independence
  - Definition
  - Conditional Independence
  - Independence and Expectation
  - Independence and Variance
  - Variance of Binomial Random Variable

Independence

- Definition:

If

$$\mathbf{P}(A|B) = \mathbf{P}(A)$$

We say that A is independent of B

- Intuition
  - Remember that conditional probability encodes our "updated belief" about the likelihood of an event IF we have "additional partial information" about the results of the random experiment.
  - If A is independent of B, occurrence of B provides no new information about occurrence of A.
  - If A is independent of B, our beliefs are not "updated" about the likelihood of A if we have the additional information that B will occur.

Independence

- An Equivalent Definition:

If

$$\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B)$$

We say that A is independent of B

Independence of Random Variables

- Independence of 2 Events:

$$\mathbf{P}(A \cap B) = \mathbf{P}(A)\mathbf{P}(B), \qquad \mathbf{P}(A|B) = \mathbf{P}(A)$$

- Independence of a RV and an Event:

$$\mathbf{P}(X = x \text{ and } A) = \mathbf{P}(X = x)\mathbf{P}(A), \qquad \forall x$$

$$p_{X|A}(x) = p_X(x), \forall x \quad \Longrightarrow \quad \mathbf{P}(A|X = x) = \mathbf{P}(A), \forall x$$

- Independence of two RVs:

$$\mathbf{P}(X = x \text{ and } Y = y) = \mathbf{P}(X = x)\mathbf{P}(Y = y), \qquad \forall x, y$$

$$p_{X|Y}(x|y) = p_X(x), \qquad p_{Y|X}(y|x) = p_Y(y)$$

$$p_{X,Y}(x, y) = p_X(x)p_Y(y)$$

Independence of Random Variables

- If two RVs X, Y are independent, then any $f(X), g(Y)$ are also independent.

- Independence of more than two RVs:
  - RVs X, Y, Z are independent if:

$$p_{X,Y,Z}(x, y, z) = p_X(x)p_Y(y)p_Z(z), \qquad \forall x, y, z$$

- Bonus Material (finer points that you may safely forget):
  - If RVs X, Y, Z are independent, then any $f(X), g(Y), h(Z)$ are also independent
  - Similarly, any $g(X, Y)$ and $h(Z)$ are also independent.
  - However, RVs of the form $g(X, Y)$ and $h(Y, Z)$ are not independent since they are both affected by Y

Example: Independence of Random Variables

Y = second roll				
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = first roll

- Are X and Y independent?

Example: Independence of Random Variables

Y = second roll				
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = first roll

- Are X and Y independent?
- No.

$$p_X(1) = \frac{3}{20}$$

$$p_{X|Y}(1|1) = 0$$

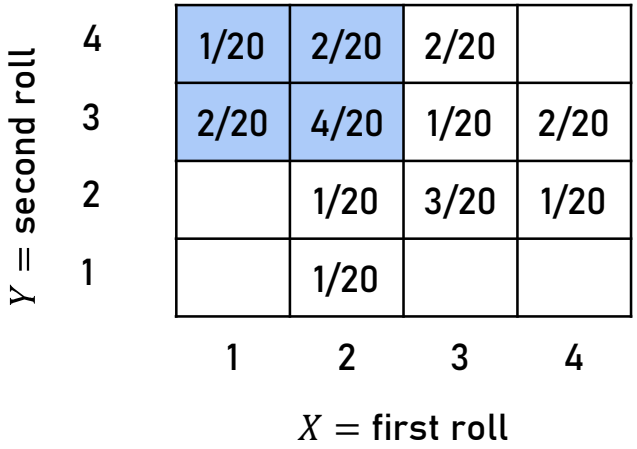
Conditional Independence of Random Variables

- Two random variables X and Y are said to be conditionally independent, given a positive probability event A if

$$\mathbf{P}(X = x, Y = y \mid A) = \mathbf{P}(X = x \mid A)\mathbf{P}(Y = y \mid A), \qquad \text{for all } x \text{ and } y,$$

$$p_{X,Y|A}(x, y) = p_{X|A}(x)p_{Y|A}(y), \qquad \text{for all } x \text{ and } y.$$

Example: Independence of Random Variables



- Are X and Y independent?
  - No.

$$p_X(1) = \frac{3}{20}$$

$$p_{X|Y}(1|1) = 0$$

- What if we condition on $X \leq 2$ and $Y \geq 3$

Example: Independence of Random Variables

Y = second roll				
4	1/20	2/20	2/20	
3	2/20	4/20	1/20	2/20
2		1/20	3/20	1/20
1		1/20		
	1	2	3	4

X = first roll

- Are X and Y independent?
  - No.

$$p_X(1) = \frac{3}{20}$$

$$p_{X|Y}(1|1) = 0$$

- What if we condition on $X \leq 2$ and $Y \geq 3$
  - Yes!

Y = second roll			
4	1/9	2/9	
3	2/9	4/9	
	1	2	

X = first roll

Independence and Expectation

- In general: $\mathbf{E}[g(X,Y)] \neq g(\mathbf{E}[X],\mathbf{E}[Y])$
- Always true: $\mathbf{E}[aX+b] = a\mathbf{E}[X]+b,$
  $\mathbf{E}[X+Y+Z] = \mathbf{E}[X]+\mathbf{E}[Y]+\mathbf{E}[Z]$

If X, Y are **independent**:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

then any $g(X)$ and $h(Y)$ are also independent:

$$\mathbf{E}[g(X)h(Y)] = \mathbf{E}[g(X)]\mathbf{E}[h(Y)]$$

Independence and Expectation

- If X, Y are independent:

$$\mathbf{E}[XY] = \mathbf{E}[X]\mathbf{E}[Y]$$

- Proof:
  - Let $g(x, y) = xy$

$$E[g(X, Y)]$$

$$= \sum_x \sum_y (xy) p_{X,Y}(x, y)$$

$$= \sum_x \sum_y (xy) p_X(x) p_Y(y)$$

$$= \sum_x x p_X(x) \sum_y y p_Y(y)$$

$$= \mathbf{E}[X]\mathbf{E}[Y]$$

Independence and Variance

- Always true: $\text{var}(aX + b) = a^2\text{var}(X)$
  $\text{var}(X + a) = \text{var}(X)$

- In general: $\text{var}(X + Y) \neq \text{var}(X) + \text{var}(Y)$

If X, Y are **independent**:

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

Independence and Variance

If X, Y are **independent**:

$$\text{var}(X+Y) = \text{var}(X) + \text{var}(Y)$$

Consider now the sum $X + Y$ of two independent random variables X and Y, and let us calculate its variance. Since the variance of a random variable is unchanged when the random variable is shifted by a constant, it is convenient to work with the zero-mean random variables $\tilde{X} = X - \mathbf{E}[X]$ and $\tilde{Y} = Y - \mathbf{E}[Y]$. We have

$$\begin{aligned}
\text{var}(X+Y) &= \text{var}(\tilde{X}+\tilde{Y}) \\
&= \mathbf{E}\big[(\tilde{X}+\tilde{Y})^2\big] \\
&= \mathbf{E}[\tilde{X}^2 + 2\tilde{X}\,\tilde{Y} + \tilde{Y}^2] \\
&= \mathbf{E}[\tilde{X}^2] + 2\mathbf{E}[\tilde{X}\,\tilde{Y}] + \mathbf{E}[\tilde{Y}^2] \\
&= \mathbf{E}[\tilde{X}^2] + \mathbf{E}[\tilde{Y}^2] \\
&= \text{var}(\tilde{X}) + \text{var}(\tilde{Y}) \\
&= \text{var}(X) + \text{var}(Y).
\end{aligned}$$

Independence and Variance

- Always true: $\text{var}(aX + b) = a^2\text{var}(X)$
  $\text{var}(X + a) = \text{var}(X)$

- In general: $\text{var}(X + Y) \neq \text{var}(X) + \text{var}(Y)$

If X, Y are **independent**:

$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

- Examples:
  - If $X = Y$: $\text{var}(X + Y) = \text{var}(2X) = 4\text{var}(X)$
  - If $X = -Y$: $\text{var}(X + Y) = \text{var}(0) = 0$
  - If X, Y are independent:
    $\text{var}(X - 3Y) = \text{var}(X) + \text{var}(-3Y) = \text{var}(X) + 9\text{var}(Y)$

Independence and Variance

- Always true: $\text{var}(aX + b) = a^2\text{var}(X)$
  $\text{var}(X + a) = \text{var}(X)$

- In general: $\text{var}(X + Y) \neq \text{var}(X) + \text{var}(Y)$

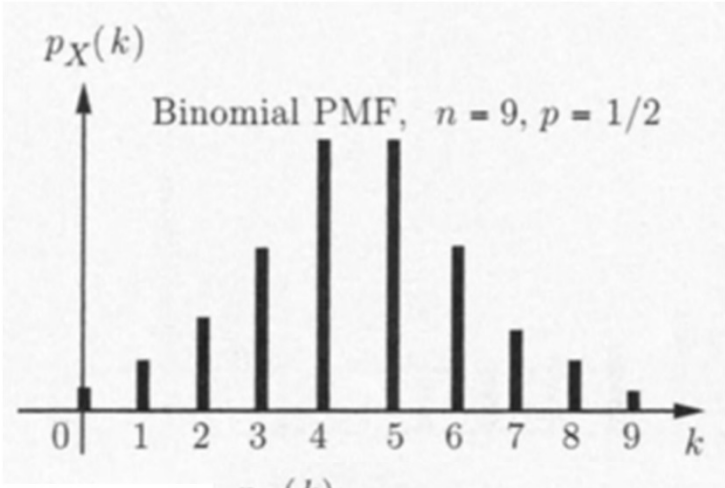
If X, Y are **independent**:

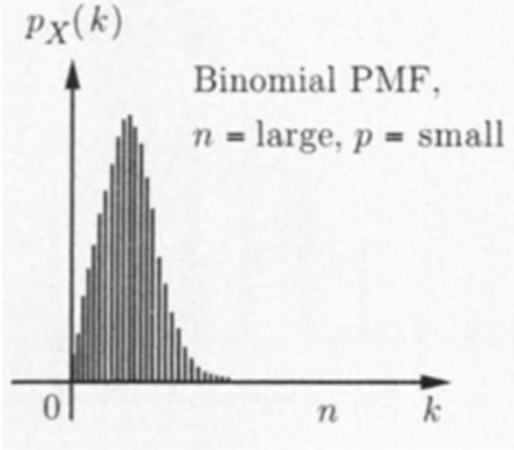
$$\text{var}(X + Y) = \text{var}(X) + \text{var}(Y)$$

- Examples:
  - If $X = Y$: $\text{var}(X + Y) = \text{var}(2X) = 4\text{var}(X)$
  - If $X = -Y$: $\text{var}(X + Y) = \text{var}(0) = 0$
  - If X, Y are independent:
    $\text{var}(X - 3Y) = \text{var}(X) + \text{var}(-3Y) = \text{var}(X) + 9\text{var}(Y)$

Binomial Random Variable

- Experiment
  - n independent tosses of a coin where at each toss, the coin comes up a head with probability p and a tail with probability $(1 - p)$.
- Sample Space
  - Set of Head (H) and Tail (T) sequences of length n
- Random Variable
  - The number of Heads in the n-toss sequence
- PMF
  - $\boldsymbol{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \ldots, n}$
- Model of:
  - Number of successes in a given number of independent trials
- Parameters: $\boldsymbol{n \in \mathbb{Z}^{++}}$, $\boldsymbol{p \in [0, 1]}$





Mean of a Binomial Random Variable

- Random Variable
  - The number of Heads in the n-toss sequence
- PMF

$$\boldsymbol{p_X(k) = \binom{n}{k} p^k (1-p)^{n-k} \quad , k = 0, 1, 2, \dots, n}$$

- Mean

$$\boldsymbol{\mathrm{E}[X] = \sum_{k=0}^{n} k p_X(k) = \sum_{k=0}^{n} k \binom{n}{k} p^k (1-p)^{n-k}}$$

  - An alternative approach:

Lets consider n random variables $X_1, X_2, \dots, X_n$ where

$$X_i = \begin{cases} 1, & \text{if } i^{\text{th}} \text{ trial is a success} \\ 0, & \text{otherwise} \end{cases}$$

$X_1, X_2, \dots, X_n$ are Bernoulli random variables with common mean p
Their sum

$$X = X_1 + X_2 + \cdots + X_n$$

is a binomial random variable with parameters n and p.

$$\mathrm{var}(X) = \mathrm{var}(X_1) + \mathrm{var}(X_2) + \cdots + \mathrm{var}(X_n)$$

$$\mathrm{var}(X) = n\mathrm{var}(X_1) = np(1-p)$$